

SUBHANKER KUMAR - PORTFOLIO KNOWLEDGE BASE

Structured reference document for an AI portfolio / personal knowledge database. Prepared from the uploaded resume, the current CGPA supplied in the request, and the GitHub repositories verified during preparation.

1. Profile

Name	Subhanker Kumar
Email	subhanker264@gmail.com
Phone	+91 7970373070
GitHub	github.com/penguin-svg
Degree	Bachelor of Technology - Computer Science & Engineering (AI/ML)
Institution	School of Information Technology, RGPV Bhopal, Madhya Pradesh, India
Graduation	2027
Current CGPA	6.76

Career / Technical Identity

Computer Science & Engineering (AI/ML) undergraduate with a project portfolio spanning machine learning, deep learning, NLP, AI systems, networking, and C++ systems programming. The portfolio demonstrates both model-oriented work and software/system implementation work.

2. Education

School of Information Technology, RGPV Bhopal - Bhopal, MP, India
B.Tech - Computer Science & Engineering (AI/ML), 2023-2027
Current CGPA: **6.76**

Delhi Public School, Gaya - Gaya, Bihar, India
Intermediate - Physics, Chemistry, Mathematics, 2020-2022

3. Knowledge Base - Technical Skills

Area	Knowledge / Technologies
Programming Languages	C, C++, Python
Core CS	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Computer Networks (CN), Databases
Python / Data	NumPy, Pandas, Matplotlib, Seaborn
Machine Learning	Supervised Learning, Unsupervised Learning, Isolation Forest, PCA, Feature Engineering, Anomaly Detection
Deep Learning	Artificial Neural Networks (ANN), CNN, RNN, LSTM
NLP / AI	NLTK, LLMs, Prompt Chaining, Retrieval-Augmented Generation (RAG), Agentic AI
Frameworks	Scikit-learn, Streamlit
Database	MySQL
Tools	Git, GitHub, VS Code
Other Project Technologies	PyTorch, Torchvision, ResNet18, Grad-CAM, Plotly, libpcap, TCP/IP, TLS, PCAP, Multithreading

4. Project Knowledge Base

Zero-Day Anomaly Detection Intrusion Detection System

Technology / Stack: Python, Scikit-learn, Streamlit, Plotly, PCA, Isolation Forest

- An anomaly-based Intrusion Detection System designed to detect zero-day cyber attacks on the NSL-KDD dataset without requiring labeled training data.
- End-to-end machine learning pipeline covering preprocessing, Mutual Information-based feature selection, PCA for dimensionality reduction, and model evaluation.
- Reported evaluation metrics in the resume: 97.8% accuracy and 94.8% F1-score.
- Interactive Streamlit dashboard with Plotly visualizations for anomaly monitoring, performance analysis, feature insights, and network-traffic exploration.

Repository: github.com/penguin-svg/Axion-ai

Satellite Imagery Land-Use Classification using CNNs

Technology / Stack: Python, PyTorch, Torchvision, ResNet18, Scikit-learn, Grad-CAM, Streamlit

- Deep learning pipeline for classifying 27,000+ Sentinel-2 satellite images across 10 land-use/land-cover classes.
- Implemented and evaluated two CNN approaches: a custom 3-4 block CNN trained from scratch and ResNet18 transfer learning pretrained on ImageNet.
- Applied data augmentation, normalization, stratified train/validation/test splitting, Adam optimization, and learning-rate scheduling.
- Evaluated accuracy, precision, recall, F1-score, and 10-class confusion matrices for quantitative model comparison.
- Integrated Grad-CAM to generate class-activation heatmaps and deployed the inference pipeline through a Streamlit web application.

DPI Engine - Offline Deep Packet Inspection System

Technology / Stack: C++17, libpcap, TCP/IP, TLS, Multithreading, PCAP

- Offline DPI engine for Ethernet PCAP analysis with IPv4 and TCP/UDP parsing, bidirectional flow tracking, and protocol-aware traffic inspection.
- TLS ClientHello SNI and HTTP Host extraction for classifying 6+ common applications/domains, including YouTube, Facebook, TikTok, GitHub, and Google.
- Configurable rule engine with three blocking policies: application, source IPv4, and domain; generates filtered PCAP files by dropping matching traffic.
- Single-threaded and multithreaded pipelines using thread-safe queues, configurable worker threads, and modular packet parsing, flow management, protocol extraction, and rule evaluation components.

Repository: github.com/penguin-svg/DPI

Multimodal Fake News Detection System

Technology / Stack: Python, NLP / ML, TF-IDF, feature engineering

- Repository identified on the user's GitHub as Fake-news-Detection-system.
- The user's project context describes a multimodal fake-news direction and a dataset with 44,898 samples: 21,417 real and 23,481 fake items.
- Dataset fields documented in the project context: title, text, subject, date, label.
- The available source material supports text preprocessing and linguistic-feature extraction as part of the project workflow.

Repository: github.com/penguin-svg/Fake-news-Detection-system

E-mail / SMS Spam Filter

Technology / Stack: Python, NLP / text classification, Streamlit

- GitHub repository identified as E-mail-SMS-SPAM-filter.
- The user's project context describes a Streamlit-based spam-classifier application using serialized model/vectorizer files.

Repository: github.com/penguin-svg/E-mail-SMS-SPAM-filter

Creditwise Loan Approval System

Technology / Stack: Machine Learning / Python

- GitHub repository identified as Creditwise-Loan-Approval-System.
- The repository title indicates a loan-approval / credit-decision project; detailed implementation claims are not expanded here beyond what is grounded in the available source information.

Repository: github.com/penguin-svg/Creditwise-Loan-Approval-System

Resume Parser

Technology / Stack: Repository project

- GitHub repository identified as Resume-Parser.
- Detailed implementation information was not available in the verified source material used for this document, so no additional technical claims are added.

Repository: github.com/penguin-svg/Resume-Parser

PortFolio AI

Technology / Stack: Portfolio application / AI

- GitHub repository identified as PortFolio_AI.
- Repository description: an AI portfolio that lets visitors ask questions about the developer.
- Repository metadata verified a public deployment homepage: portfolio-ai-p0jg.onrender.com.

Repository: github.com/penguin-svg/PortFolio_AI

PortFolio-AI

Technology / Stack: Portfolio / AI repository

- A second GitHub repository named PortFolio-AI is present in the account.
- This is listed separately because it is a distinct repository from PortFolio_AI.

Repository: github.com/penguin-svg/PortFolio-AI

5. Knowledge Areas for Portfolio Q&A;

Programming & Problem Solving: C/C++, Python, DSA, arrays, trees, graphs, dynamic programming, backtracking, OOP

Machine Learning: Supervised and unsupervised learning, feature engineering, PCA, Isolation Forest, anomaly detection, model evaluation

Deep Learning: ANN, CNN, RNN, LSTM, PyTorch, transfer learning, ResNet18, data augmentation, Grad-CAM

NLP / Text AI: NLTK, TF-IDF, text preprocessing, spam classification, fake-news detection

Generative / Modern AI: LLMs, prompt chaining, RAG, agentic AI

Networking & Systems: TCP/IP, Ethernet, PCAP, IPv4, TCP/UDP, TLS, SNI, HTTP Host, deep packet inspection, multithreading

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Plotly, Streamlit

Databases & CS Fundamentals: MySQL, DBMS, OS, CN, OOP

Developer Workflow: Git, GitHub, VS Code

6. Achievements

- Solved 250+ Data Structures and Algorithms problems on LeetCode, with demonstrated practice across arrays, trees, graphs, dynamic programming, and backtracking.

7. GitHub Portfolio Index

Repository	Known role in portfolio
Fake-news-Detection-system	Fake-news / NLP / ML project repository
Resume-Parser	Resume parsing project repository
PortFolio_AI	AI portfolio that supports question answering about the developer
DPI	Deep Packet Inspection system
Creditwise-Loan-Approval-System	Loan approval / credit project
PortFolio-AI	Second portfolio AI repository
E-mail-SMS-SPAM-filter	Email/SMS spam filtering project
Axion-ai	Zero-day anomaly detection / IDS project

8. Source & Accuracy Notes

This document is intended as a portfolio knowledge base rather than a replacement for the resume. Resume-derived details include the education, technical skills, three detailed projects, and LeetCode achievement. The CGPA of 6.76 was supplied in the current request. GitHub-derived entries were limited to repositories and metadata that could be verified. Where a repository's implementation details were not available from the verified source material, the document explicitly avoids inventing technical claims.